Lab Assignment 1: Checklist & Recommendations

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This document aims to provide students with a way to check the quality of their lab work. To do so, it includes a Checklist section detailing the expected outcomes of the bulletin's activities as well as an Example of how to properly answer in a justified manner.

Be reminded that you may write answers to questions directly on your notebook using Markdown (text) cells.

Checklist

By the end of lab assignment 1, each student pair should have:

- Activities 1 to 5, in notebook `puzzle-p2`
  - Function implementing each of the described heuristics
  - Instructions registering all new heuristics so that they are ready to be executed
- Activity 6, in notebook `puzzle-p2`
  - Output showing the results of running all search algorithms (including new heuristics)
  - For each new heuristic, a written statement indicating whether they are Admissible and/or Consistent. This statement should be based on the output of your code.
- Activity 7, in notebook `alg-compare-p3`
  - A writen statement justifying why the two chosen heuristics are "the best two" for you. This statement should reference the experiments carried out in notebook `puzzle-p2`.
  - A deeper analysis and comparison of the two chosen search algorithms, following the example of notebook `alg-compare-p3`. This analysis shoud feature:
    * The code to setup and execute the experiment, gather the results, and print the statistics.
    * A written statement comparing the output of the experiments with regards to the following metrics: Solution quality (cost) and Computational costs (time and memory consumptions).

Example of a justified answer

To justify an answer, your written statement must explicitly mention the output data of your experiments. Both the output and the code that produced such an output should be available in the notebook.

For instance, for the question:

*-Is the heuristic 2*Manhattan Distance an Admissible and/or Consistent heuristic?*

Given the following code output, a justified answer could be:

Output:

```
New Heuristic (2*Manhattan)
----------------------------------------
H value in the initial state: 32
H value in the goal state: 0
BFS optimal cost: 22
Is H non-negative in this example? True
Does H equal to 0 in the goal state? True
Does H never exceed the optimal cost? False
Consistency test results:
{'consistent': False, 'complete': False, 'states_checked': 1,'counterexample
    ': {'state': [7, 8, 1, 4, 0, 6, 2, 3, 5], 'child': [7, 0, 1, 4, 8, 6, 2,
     3, 5], 'h_state': 32, 'h_child': 30, 'step_cost': 1}}
```

Answer: "We have tested the 2*Manhattan Distance heuristic over an initial state of the 3x3 puzzle and its estimated cost was of 32. However, that same state was solved optimally in 22 steps. This result certifies that this heuristic overestimates the cost and therefore is not Admissible and, thus, not Consistent either."